

ASHIM PAUDEL

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EDUCATION

MS in Computer Science | Louisiana State University | Expected Dec 2026

GPA: 4.0 | Relevant Courses: Machine Learning, Deep Learning, Applied LLM Development, Computational Chemistry

BS in Computer Science | Mississippi State University | May 2022

GPA: 3.41 | Relevant Courses: DBMS, Programming Languages, AI Robotics, Algorithms, Microprocessors

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, MATLAB, Bash

Frameworks/Libraries: PyTorch, TensorFlow, Scikit-Learn, Transformers, OpenCV, NumPy, Pandas

Agentic Workflows: LangChain, LangGraph, MCP, RAG, FastAPI

Database/Technologies: MySQL, SparkSQL, Apache Kafka, Apache Hadoop, Git, Docker

AI Tools: Claude Code, Cursor, Codex, Copilot

Scientific Computing: NAMD, GROMACS, VMD, HPC, SLURM

EXPERIENCE

Graduate Research Assistant

Jan 2025 - Present

Louisiana State University | Baton Rouge, LA

- Architected a graph neural network (GNN) active learning framework with MARTINI coarse-grained molecular graphs and query-by-committee sampling to predict polymer-drug encapsulation, achieving $R^2 = 0.88$ with 45% fewer GROMACS simulations than uniform random sampling.
- Developed AI agents to automate molecular dynamics workflows, including molecular structure parsing, simulation setup, execution, and trajectory analysis.
- Developed an LLM-assisted scientific literature extraction pipeline using prompt engineering to extract and structure polymer and molecular simulation data from research publications.
- Applied spaCy NLP to extract polymer and force-field mentions from scientific papers, detecting $\sim 8\times$ more polymer name variants than regex parsing.
- Performed molecular dynamics simulations of polymer-drug systems, including topology generation, system solvation, and trajectory analysis; used Bash scripts to automate multi-step workflows and Slurm for HPC job execution.

Software Engineer

Jul 2022 - Jan 2024

Orbbec | Troy, MI

- Developed a real-time image processing pipeline in C++ for converting RGB-D images into stereoscopic view. Integrated Orbbec's proprietary algorithm, leveraging OpenCV and Boost for efficient multi-threaded data handling, and used StereoKit for XR rendering across VR platforms.
- Developed 'The Eye for AIs,' an advanced image processing application that isolates objects, computes volumes, and performs detailed analysis using color and depth images from a depth camera. Leveraged vision language models (VLMs) and large language models (LLMs) to segment and interpret regions of interest from images.
- Led integration of a tailored LLM API into a customer chatbot application. Improved technical response accuracy by 18% by fine-tuning the LLM model using the company software development kit, product specifications, and API documentation.
- Led the design of an iOS prototype to capture and transmit synchronized RGB camera and LiDAR depth sensor data over the network to perform real-time stereoscopic 3D reconstruction. Conducted in-depth research on Apple's LiDAR sensor and depth APIs.
- Collaborated with Nvidia to evaluate the compatibility of their body and hand-tracking technologies with the Orin and Jetson Nano platforms. Conducted thorough testing of Orbbec SDKs, documenting test cases, outcomes, and errors.

PUBLICATIONS

- Paudel, A., Shakya, A., et al. A structured dataset of polymers in molecular dynamics simulations extracted by LLM-assisted text mining. Scientific Data (2026). Accepted.